

A Fixed P3b Endpoint Carries Over to an Independent Auditory Oddball Dataset: A Preregistered Cross-Paradigm Robustness Test

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October 9, 2026

Version 0.1.0 Preprint, not peer reviewed

Abstract

The P3b is among the most studied event-related potentials, but a measurement chosen after seeing data can find an effect almost anywhere on the scalp. We therefore fixed one endpoint in advance, the mean voltage at Pz from 300 to 600 ms after stimulus onset recommended by ERP CORE [1], with its contrast direction, participant-level unit of inference, artifact threshold and stopping rule, before inspecting any amplitude in an independent dataset: the auditory three-stimulus oddball task of OpenNeuro ds003061 [2]. All 13 participants showed a positive correct-target-minus-correct-standard contrast; the mean was $+5.65 \mu\text{V}$ (95% interval 4.83 to 6.48), $t(12) = 15.00$, two-sided $p = 3.9 \times 10^{-9}$, Cohen’s $d_z = 4.16$, unchanged across three artifact thresholds. Recomputing every statistic from the 13 stored participant contrasts reproduces the report exactly and adds small-sample quantities: an exact sign-flip $p = 2.4 \times 10^{-4}$, a noncentral- t interval for d_z of 2.42 to 5.88, and a 95% prediction interval for a new participant of $+2.58$ to $+8.73 \mu\text{V}$. This is cross-paradigm robustness evidence, not a direct replication.

1 Question

Does the classic P3b target enhancement survive when the measurement is fixed in advance and applied to an independent public EEG dataset with a different sensory modality? ERP CORE used an active visual oddball task; the external dataset used an active auditory oddball task [3]. The test is therefore a robustness check of a fixed endpoint, not a literal replication.

2 Preregistered design

The endpoint was frozen before any EEG amplitude was inspected: electrode Pz, mean voltage 300 to 600 ms post-stimulus, contrast correct targets minus correct standards, one contrast per participant (trials are pooled within participant, never treated as independent people), a $150 \mu\text{V}$ peak-to-peak artifact rule, and a minimum of 20 correct targets and 100 correct standards per run. The dataset metadata was audited at a pinned commit without inspecting signals: all 39 channel tables contain Pz, sampling is 256 Hz throughout, and there are three runs per participant. One visibly truncated recording (sub-012 run 1) failed the frozen minimum before any signal analysis and was excluded; the participant remained through runs 2 and 3. The 38 eligible runs contributed 3,606 accepted target and 19,481 accepted standard epochs.

3 Confirmatory result

Quantity	Estimate
Participants	13
Mean target-minus-standard contrast	+5.654 μV
Median contrast	+5.570 μV
95% interval for the mean	+4.833 to +6.476 μV
Two-sided one-sample test	$t(12) = 15.004, p = 3.88 \times 10^{-9}$
Cohen’s d_z	4.161
Positive participant contrasts	13/13 (range +3.22 to +8.72 μV)
Wilcoxon sensitivity	$W = 0, p = 0.000244$

Table 1: Preregistered confirmatory result at Pz, 300–600 ms.

The prespecified artifact-threshold sensitivity leaves the conclusion unchanged: 100 μV gives +5.827 (4.893 to 6.761); 150 μV (confirmatory) +5.654; 200 μV +5.738 (4.825 to 6.652), with all 13 participants retained in each.

4 Post-hoc recomputation from participant contrasts

Because the endpoint is one number per participant, every reported statistic can be recomputed exactly from the 13 stored contrasts, without the raw EEG. We did so as an independent check, and added quantities suited to $n = 13$. Not preregistered; all reported.

Quantity	Value
Largest difference from the reported mean, t , d_z and interval	0
Hedges’ small-sample g_z [4]	3.896
Noncentral- t 95% interval for d_z	2.424 to 5.883
Exact sign-flip permutation p (two-sided, 2^{13} patterns)	0.000244
Leave-one-participant-out mean range	+5.399 to +5.858 μV
Leave-one-out smallest t / largest p	13.77 / 2.8×10^{-8}
95% prediction interval, a new participant	+2.58 to +8.73 μV

Table 2: Recomputation from the participant-level contrasts (`results/summary.json`).

5 Discussion and limits

The effect is large, present in every participant, and not driven by any single person or artifact threshold. The prediction interval excludes zero under a normality assumption, a statement about a new draw from this population rather than about all auditory oddball studies. What the result supports is narrow: a fixed posterior 300–600 ms target enhancement appears in an independent, small auditory dataset despite a change of modality and paradigm, which a post-hoc search of the scalp could not demonstrate. It does not show that visual and auditory oddball tasks engage identical processes, localize a generator, or support any diagnostic or individual-prediction use. The sample is 13 people. The recomputation here starts from the stored participant contrasts; it did not re-run signal processing from the raw EEG, whose download and preprocessing are specified in the repository and verified there by checksums.

Reproducibility and AI assistance

Code, protocol, metadata audit, results and tests:

<https://github.com/lindgreendavid/neuro-signal-lab>

Interactive laboratory:

<https://lindgreendavid.github.io/neuro-signal-lab/>

The analysis code, tests and drafts of this text were produced with substantial assistance from an AI system (Claude, Anthropic); the author is responsible for the content.

References

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